Retrieving Behavioral Equity Peers from Filing Text and Return History

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Abstract

We study quarterly behavioral equity-peer retrieval. Given a focal company, we rank five firms expected to co-move with it during the following quarter. Co-movement is measured after removing Fama–French five-factor (FF5) exposures. We test whether annual Form 10-K Item 1 business text available before ranking adds information beyond recent return co-movement, and whether any gain changes with additional return history. Our training-free baseline uses the standardized residual-return time series itself as each security's embedding. Pairwise cosine similarity over shared dates exactly equals sample residual correlation.

Six walk-forward tests use one quarter to rank firms and the next to evaluate them, with 1,614–1,631 securities per test. The return path attains mean normalized discounted cumulative gain at five (NDCG@5) of 0.336, compared with 0.300 for term-frequency–inverse-document-frequency (TF-IDF) and 0.279 for the dense Granite-r2 text encoder. Filing similarity and residual co-movement are nearly unrelated at the pairwise level: across 1,301,691 pairs, four dense filing encoders have Spearman correlations of only 0.029–0.046 in the same quarter and 0.027–0.049 in the next. Even so, an 80/20 combination of return- and text-based ranks, fixed after selection on twelve earlier tests, reaches 0.373 and improves on the return path in all six. With four quarters of return history, text no longer improves the ranking. Across 19 transitions spanning 2020–2026, the return path leads TF-IDF in 18.

For the availability audit, each five-name list is fixed before following-quarter returns are known; missing candidates are not replaced. Of cases with the required inputs at ranking time, 99.2–99.6% remain evaluable. In this setting, filing text offers limited help in generating candidates. The experiment does not test valuation-peer selection, hedge construction, trading, or analyst outcomes, and it still requires recent return history.

CCS Concepts

• **Information systems** → **Retrieval models and ranking**; • **Applied computing** → *Economics*.

Keywords

financial embeddings, return correlations, peer retrieval, multi-modal retrieval, financial benchmark construction

ACM Reference Format:

Anonymous Author(s). 2026. Retrieving Behavioral Equity Peers from Filing Text and Return History. In *Proceedings of 7th ACM International Conference on AI in Finance (ICAIF '26).* ACM, New York, NY, USA, 8 pages.

1 Introduction

We study *behavioral equity-peer retrieval*: for a focal company, rank firms expected to co-move with it in the next quarter after both firms' returns have been adjusted for the Fama–French five-factor model (FF5). We call the returns left after this regression *residual returns* and their correlation *behavioral similarity*. One use is to generate candidates for an equity-risk analyst investigating shared shocks not captured by FF5 [7]. The output is a candidate list; valuation-peer selection, executable hedges, and trading signals fall outside the task.

Prior work measures company similarity from 10-K business descriptions using lexical methods or pre-trained language-model embeddings [22, 34]. We use these representations without fitting them to behavioral-peer or return labels and retrieve their nearest neighbors. We ask whether similar filing language predicts similar residual returns. Similar businesses can respond differently, and firms exposed to a shared return driver omitted from FF5 can co-move even when their filings do not describe the connection.

Our return comparator is each security's standardized residual-return path over common dates. Equal norms make cosine similarity and the dot product divided by the number of dates exactly equal to sample residual correlation. We call this vector indexed by date a *return-path embedding*. The identity reproduces the correlation ranking for the same period. Predictive value must be measured in the following period, so we test whether the ranking persists and how it compares with filing-derived neighbors.

Each test uses two consecutive periods. In a *walk-forward* (WF) transition, feature period W_t supplies the ranking and target period W_{t+1} evaluates it. We form the ranking after W_t closes and before observing any W_{t+1} outcome; *feature-time* refers to the information and eligibility available at that point. A *vintage* filing was public before W_t began. For comparison, a *same-window* (SW) diagnostic ranks and evaluates within a single period. NDCG@5 measures ranking quality among the first five results and is defined in §3.3; *text–return alignment* is pairwise Spearman rank correlation ρ_S.

The contribution is a leakage-controlled benchmark for testing whether vintage filing text adds information beyond recent residual returns when retrieving peers for a subsequent period:

(1) *Return baseline.* Each security's parameter-free return path can be indexed independently. Its nearest-neighbor ranking exactly matches the correlation ranking for the same period (§4.1).
(2) *Time alignment.* The retrospective benchmark uses vintage 10-K inputs and selects configurations from earlier windows only. A separate audit freezes each five-name list before target-period returns are available, so later availability cannot change its candidates.
(3) *Pairwise disconnect.* Across 1.3 million pairs, four dense filing encoders are nearly unrelated in rank to same- or next-quarter residual co-movement ($\rho_S \leq 0.049$). Return paths are stronger on their own; adding text helps only when return histories are short.

ICAIF '26, Milan, Italy
2026.

This setup estimates the incremental value of vintage filing text over a return baseline. Models trained on behavioral-peer or return labels address a different learning problem.

2 Related Work

2.1 Text-Based Financial Peers

Hoberg and Phillips [21, 22] measure product-market relationships via 10-K product-description similarity. Lee et al. [28] exploit investor co-searches, and Bhojraj and Lee [6] select valuation peers through warranted multiples. We instead ask whether filing-derived and industry similarity can retrieve firms that will co-move over a short horizon, a target this literature does not typically study. Box [8] links firm-specific Reuters news similarity to future cross-firm return co-movement. Dyer et al. [11] establish a similar association for corporate disclosures. Our test asks whether filing-derived similarity can replace recent return history when FF5 residual correlation in the following period defines the retrieval target. Filing-embedding neighbors have also been associated with financial metrics, including return correlation [34]. Bonne et al. [7] combine returns and risk signals for peer grouping, while SimStock learns peer vectors designed to remain stable over time [23]. Dolphin et al. [10] learn contrastive asset embeddings from stocks' repeated appearances among one another's most-correlated raw-return neighbors across rolling periods. Their downstream tests cover sector classification and hedging. Our FF5-residual path reproduces each period's residual-correlation ranking without training and tests whether that ranking persists into the next period. Feng et al. [16] learn contextual time-series embeddings and validate them on Global Industry Classification Standard (GICS) labels and portfolio construction. A closely related recent preprint is Future-Aligned Soft Contrastive Learning (FASCL) [29]. It trains a Transformer against future return correlations and evaluates 4,229 equities against thirteen statistical and learned baselines. Our experiments use a training-free FF5-residual representation in repeated walk-forward tests and measure whether vintage filing text adds to recent return history. FASCL outperforms historical sample correlation, so learned time-series retrieval can exceed the return-path baseline studied here. We did not have access to historically dated GICS or security-matched Text-based Network Industry Classification (TNIC) [22] panels. The filing comparison is limited to the tested off-the-shelf encoders, TF-IDF, and two-digit Standard Industrial Classification (SIC) groups. Lagged annual TNIC is left for future work.

2.2 Behavioral Versus Semantic Embeddings

beeFormer [35] reports that off-the-shelf semantic embeddings miss behavioral interaction patterns, while MoRec [38] narrows the gap through end-to-end adaptation. We examine the same distinction in a walk-forward retrieval task whose target is residual correlation.

2.3 Correlation Structure and Machine Learning

In five-year forecasts, Elton and Gruber [12] find that simple averages outperform the full historical pairwise-correlation matrix. We examine a shorter horizon: whether each query's top-five ordering survives into the following period. RiskMetrics popularized exponentially weighted covariance forecasts [25], and Engle [13] model time-varying conditional correlations. We study a representation that can be computed independently for each firm and used in vector search, then benchmark its peer retrieval against vintage filing text. Mantegna [31] derives a hierarchical stock taxonomy from daily-return correlations. Laloux et al. [26] apply random-matrix theory to separate signal from noise in large correlation matrices. Avellaneda and Lee [3] use principal-component ("eigen-portfolio") residuals for mean-reversion strategies. These studies use correlation structure for taxonomy, denoising, or trading. Here it defines coordinates for peer retrieval. Ledoit–Wolf shrinkage [27] yields well-conditioned estimates when the number of securities far exceeds the number of dates ($N \gg T$). Low-rank spectral decompositions compress $N \times N$ matrices [20]. We use both as baselines. Feng et al. [15] also formulate a ranking problem, while Ghiye et al. [19] apply a no-lookahead graph convolutional network (GCN) to bond recommendation, another temporally ordered setting.

3 Data and Problem Formulation

3.1 Company Universe and Text Descriptions

The set of 2,200 candidate securities was fixed on April 12, 2026 and assembled retrospectively from current U.S. Securities and Exchange Commission (SEC) company metadata and Global Database of Events, Language, and Tone (GDELT) Global Knowledge Graph organization coverage [18], including a 2025–2026 news-visibility filter. Requiring a complete return vector leaves 1,921–1,995 securities per feature window. The file does not reconstruct the tradable universe at each historical date and is not a point-in-time security master. All results are conditional on this panel.

The primary text comparison uses Item 1 ('Business') text from the annual U.S. Form 10-K, available through the SEC's EDGAR filing system [33] before the start of each feature window. We call text restricted in this way *vintage-correct*, or simply *vintage*. Every text comparator in this paper uses vintage text unless stated otherwise. For a $W_t \to W_{t+1}$ transition, each list is formed after W_t closes from its returns, but the 10-K must have been public before W_t began. Requiring tickers in both return windows and a vintage 10-K yields 1,614–1,631 securities. Filing availability in W_{t+1} is not required. Because this common evaluation set requires returns from both periods, the primary NDCG analysis is retrospective. The collection pipeline caps Item 1 excerpts at 8,000 characters. We use neither generated descriptions nor descriptions published after the feature period.

Company descriptions are embedded with four dense encoders: `all-MiniLM-L12-v2` [32, 36], BAAI General Embedding (BGE; `BAAI/bge-base-en-v1.5`) [37], mean-pooled SEC-BERT (a filing-pre-trained BERT released with FiNER; `nlpaueb/sec-bert-base`) [30], and Granite-r2 [4] (IBM, ModernBERT-based, 2025).

3.2 Return Windows and Correlation Target

Adjusted-close prices from Yahoo Finance, retrieved with `yfinance` [2], are converted to daily simple returns. For every security i and date t, we subtract the risk-free return $R_{f,t}$ from the security return $R_{i,t}$ and regress the resulting daily excess return, with an intercept, on the Fama–French five-factor (FF5) model [14], using daily factors from the Kenneth R. French Data Library [17].

The factors capture market, size, value, profitability, and investment exposures. Residuals $\varepsilon_{i,t}$ remove their fitted linear contribution. We construct seven non-overlapping, approximately quarterly calendar blocks W_1–W_7 spanning May 2024 through January 2026; exact boundaries and per-window trading-day counts are given in Table 1. The windows contain N = 1,921–1,995 securities and T = 47–77 common trading dates; N varies because different subsets of the 2,200-ticker candidate panel have complete returns within each calendar interval. A security is retained only if all T returns are observed; missing returns are never imputed as zero. For each window, residuals are demeaned and divided by the population standard deviation (dividing by T, not $T-1$; equivalently, `ddof= 0` in NumPy) before all downstream computation.

The correlation target $C \in \mathbb{R}^{N\times N}$ is the Ledoit–Wolf (LW) shrinkage estimate [27] of residual covariance, converted to correlations. This analytic noise-reduction estimator pulls off-diagonal sample correlations toward zero. Its shrinkage intensity is $\delta \in [0.700, 0.787]$ across W1–W7 and $[0.62, 0.81]$ across the full 20-window extension. The wider historical range reflects larger T in those blocks. Under the standardization convention, the shrinkage target is the identity and $C_{\mathrm{LW}} = (1-\delta)C_{\mathrm{sample}} + \delta I$ exactly, where C_{sample} is the unshrunk sample correlation matrix and I is the identity matrix, so LW shrinkage preserves all off-diagonal candidate rankings (§4.1).

Evaluation protocols. The *return-path embedding* is a company's z-scored residual vector $\mathbf{x}_i \in \mathbb{R}^T$; dividing by $\sqrt{T}$ gives a unit vector whose dot product equals sample correlation (§4.1).

In the *same-window* (**SW**) diagnostic, features and relevance both come from W_t. The rank-128 spectral baseline is *transductive*: it requires the entire W_t company correlation matrix when peers are ranked. The return path and text representations can score a security from their respective observed inputs without refitting a learned encoder. In the primary *walk-forward* (**WF**) evaluation, representations come from W_t and relevance from W_{t+1} correlations. All primary modern results use W_1–W_7. Algorithm 1 gives the return-path construction.

3.3 Problem Statement and Evaluation Metric

Task. Given a query company q, retrieve the top-k companies (k=5 for primary results) whose residual returns co-move most with q (single-company peer retrieval). We fix $k = 5$ as a manageable review list and have not optimized it for a production workflow. At quarter end, an equity-risk analyst could inspect the five candidates for possible shared non-factor drivers. The system produces neither a driver label nor an automated portfolio action.

Relevance. We use $\max(0, C_{\mathrm{LW},qi})$, where $C_{\mathrm{LW},qi}$ is the LW correlation between q and candidate i estimated over the evaluation window (distinct from the Spearman alignment coefficient ρ_S of §1). The task targets positive co-movement, so negative correlations receive zero relevance. The same transform is applied to the retrieved and ideal lists. For off-diagonal pairs, $\max(0, C_{\mathrm{LW},qi}) = (1-\delta)\max(0, C_{\mathrm{sample},qi})$; this common positive scale cancels from both the observed and ideal discounted gains, so NDCG is exactly invariant to the LW intensity. We therefore do not tune shrinkage as part of the relevance target.

Metric. Ranking quality is measured by NDCG@5 (Normalized Discounted Cumulative Gain at depth 5) [24]: NDCG@5 = DCG@5/IDCG@5. Here DCG@5 = $\sum_{i=1}^{5} \mathrm{rel}_i/\log_2(i+1)$ and rel_i is the relevance at rank i, as defined above. IDCG@5 is the DCG@5 of the ideal ranking over the eligible set, which contains every company except the query. An implementation check against this ideal ranking gives NDCG@5 = 1.000. Within each transition we average over all eligible queries, then weight the transition means equally. These are the reported macro means, and ± denotes their population standard deviation (SD). Every eligible security serves as a query: 1,614–1,631 per transition among firms with vintage text at ranking time, and 1,919–1,979 in the full-return set. No query subsampling is involved. The query itself is excluded from both lists, and score ties are broken reproducibly.

Table 1: Experimental configuration (all windows).

Parameter	Value
Primary windows (W1–W7)	
W1	2024-05-01 – 2024-08-20 (T=77)
W2	2024-08-21 – 2024-11-20 (T=65)
W3	2024-11-21 – 2025-01-31 (T=47)
W4	2025-02-03 – 2025-04-30 (T=61)
W5	2025-05-01 – 2025-07-31 (T=63)
W6	2025-08-01 – 2025-10-31 (T=65)
W7	2025-11-03 – 2026-01-28 (T=59)
Extension (H1–H13)	Feb 2020–Apr 2024; T=62–86
N (securities)	1,921–1,995 (full); 1,614–1,631 (vintage)
Factor model	FF5 + intercept
Correlation	LW ($\delta \in [0.700, 0.787]$ W1–W7; $[0.62, 0.81]$ full)
Standardization	Population (ddof= 0)
Queries	all (1,614–1,631 vintage; 1,919–1,979 full)
Retrieval depth	$k = 5$ primary; $\{1, 5, 10, 20\}$ sensitivity
Metric	NDCG@5, rel = $\max(0, C_{\mathrm{LW},qi})$
TF-IDF	50,000 (1,2)-gram features; selected on H1–H13

4 Methods

We compare three families of methods. The *return-path embedding* (§4.1) maps each security to its z-scored residual-return vector and retrieves neighbors by cosine similarity or the equivalent dot product after unit normalization. It has no learned parameters. Same-window NDCG@5 is 1.000 by algebraic identity and serves only as an implementation check. A *spectral baseline* (§4.2) compresses the full Ledoit–Wolf correlation matrix into rank-128 eigenvectors. This transductive method tests low-rank compression. The *text and industry* group consists of four frozen dense encoders scored by cosine similarity, a lexical TF-IDF baseline scored the same way, and two-digit SIC codes (SIC-2) as a coarse industry comparator. All primary text inputs are 10-K Item 1 passages cut off before each window. A deterministically seeded uniform-random ranking measures chance performance.

4.1 Return-Path Embedding

For window W, define the map $\phi_W(i) = \mathbf{x}_i/\sqrt{T} \in \mathbb{R}^T$, where $\mathbf{x}_i$ is company i's z-scored residual path on the window's shared dates. The map has unit norm, $\|\phi_W(i)\|_2 = 1$, and

$$\phi_W(i)^\top \phi_W(j) = \cos(\phi_W(i), \phi_W(j)) = C_{\mathrm{sample},ij}. \tag{1}$$

Ledoit–Wolf shrinks off-diagonal entries uniformly ($C_{\mathrm{LW}} = (1-\delta)C_{\mathrm{sample}} + \delta I$; defined in §3.2), preserving all pairwise rankings. Inner-product retrieval on $\phi_W(i)$ therefore recovers the

Algorithm 1 Return-path embedding: construction and retrieval. No parameters are learned; the inner product in step 6 equals the sample residual correlation.

Require: Window dates $\mathcal{D}$, returns R, risk-free series R_f, FF5 factors F, securities S, depth k

1: **for** each $i \in S$ **do**
2: Regress $R_{i,t} - R_{f,t}$ on F_t with intercept over $\mathcal{D}$; retain residuals ε_i
3: $\mathbf{x}_i \leftarrow (\varepsilon_i - \bar{\varepsilon}_i)/\mathrm{std}(\varepsilon_i)$ {ddof= 0}
4: $\phi(i) \leftarrow \mathbf{x}_i/\sqrt{T}$ {unit norm}
5: **end for**
6: Index $\Phi = \{\phi(i) : i \in S\}$ in an inner-product store
7: **Query** q: return top-k by $\langle \phi(q), \phi(i) \rangle$

Ensure: Top-k peers of q

exact LW ranking. When ranking and evaluation use the same period, NDCG@5 is 1.000 by construction. Equivalently, z-scoring (ddof= 0) enforces $\|\mathbf{x}_i\|_2^2 = T$. Cosine similarity on $\mathbf{x}_i$, its dot product scaled by $1/T$, and sample correlation are identical.

The identity is a diagnostic. Prediction carries the W_t ranking into W_{t+1}. A security is mapped from its own returns and public FF5 factors without refitting an encoder. If X stacks standardized paths, then $C_{\text{sample}} = XX^\top/T$. Storing and indexing X costs $O(NT)$, permits independent insertion, and avoids materializing an $O(N^2)$ matrix. Each 32-bit floating-point (float32) vector uses 188–308 bytes here. We report one return-path row because direct LW lookup, unit-vector dot product, and cosine give the same ranking.

4.2 Spectral Baseline

The spectral baseline compresses the Ledoit–Wolf correlation matrix into its top-128 eigenvectors, each scaled by the square root of its eigenvalue $\sqrt{\lambda_k}$, and retrieves neighbors by dot product. Unlike the return-path vector, it is transductive: ranking peers requires the full C_{LW} matrix for the feature period. C_{sample} has rank at most T (here $T - 6$ due to FF5 plus intercept, yielding 41–71 across windows). Since $T \leq 77 < 128$ in every window, the rank-128 basis spans the entire sample-correlation row space plus 57–87 degenerate directions at eigenvalue δ.

5 Experiments

5.1 Filing and Industry Comparators

TF-IDF is fitted independently in every feature window, so its vocabulary and inverse-document-frequency weights use no future filing. We select its configuration from a fixed five-configuration grid using only the twelve earlier 2020–2024 transitions H1→H2 through H12→H13 (Table 1), then freeze it before the modern evaluation. The selected configuration uses 50,000 unigram-and-bigram features, minimum document frequency 2, English stop words, Unicode accent stripping, log-scaled term frequency, and per-window refitting. Dense encoders retain their published pretrained weights and encode the same Item 1 passages, using 128 tokens for MiniLM and 512 for BGE and SEC-BERT. Granite-r2 is evaluated at its 8,192-token limit over the full stored excerpt and, for comparison, with a 512-token limit; cosine similarity determines the ranking. MiniLM is included as a compact general-purpose encoder, BGE as a retrieval-oriented encoder, SEC-BERT as a domain model, and Granite-r2 as a recent long-context dense encoder. We do not fine-tune on the return target. SIC-2 assigns equal score to candidates in the query's two-digit SIC group and zero otherwise; deterministically seeded noise resolves ties.

Historically selected rank fusion. For each query, we convert similarities to within-query average percentile ranks $r_{\text{return}}, r_{\text{text}} \in [0, 1]$ (higher similarity receives a higher rank; exact ties share their average) and score candidates by $\alpha\, r_{\text{return}} + (1 - \alpha) r_{\text{text}}$. The same twelve historical transitions select $\alpha \in \{0, .2, .4, .6, .8, 1\}$; the selected $\alpha = 0.8$ and the TF-IDF configuration are both frozen before W1→W2 through W6→W7. Neither component nor its weight is selected on a modern target.

5.2 NDCG@5 Retrieval Quality

Walk-forward NDCG@5 (primary evaluation). Table 2 reports walk-forward results. On the full-return universe, the path reaches 0.339 ± 0.007; spectral dot reaches 0.336 ± 0.006, as expected because rank 128 spans the sample-correlation row space (§4.2). On the vintage universe, return path, TF-IDF, and Granite-r2 over the full stored excerpt attain 0.336 ± 0.007, 0.300 ± 0.010, and 0.279 ± 0.006; the path leads every filing-based method in 6/6 transitions. Across-transition summaries are descriptive because adjacent transitions share a window.

Combining text and returns; adding return history. The historically selected, frozen 80/20 blend reaches 0.373 ± 0.008, improving the one-window path by 0.037 in 6/6 transitions (range +0.028 to +0.050); path-minus-TF-IDF gaps range from +0.021 to +0.053. For M periods of return history ending at W_t, we concatenate paths standardized separately within each period, normalize each row, and rank by cosine. Within each transition, all $M = 1, 2, 3, 4$ comparisons use the subset eligible for all four history lengths (N = 1,556–1,628). The return paths yield 0.336/0.391/0.414/0.430 and the frozen blends yield 0.373/0.403/0.416/0.425: gains of +0.038/+0.012/+0.003/−0.005 and wins in 6/6, 6/6, 5/6, and 0/6 (Fig. 1). Separate weight selection for each M on nine common 2020–2024 transitions chooses $\alpha = 0.8$ for $M = 1, 2$ and pure returns for $M = 3, 4$. Text improves paths built from one or two periods, but its gain shrinks with longer histories and is absent with four. Among the 18 transitions with enough prior history for $M = 2$, it reaches 0.421 ± 0.028 versus 0.368 ± 0.030 for $M = 1$ and improves on $M = 1$ in all 18. These results are descriptive; the one-period path requires the least history.

Audit of lists fixed before evaluation. We separately rank all firms with returns and vintage text available at W_t (N = 1,615–1,632), after W_t closes and before observing any W_{t+1} return. We then freeze the top five. A query–method pair is dropped without reranking if the query or any selected peer lacks W_{t+1} returns. For each evaluable list, we correlate the query's raw residual path with the daily mean of its five peers' raw FF5 residual paths in W_{t+1}. Mean query–basket correlations are 0.208/0.186/0.248/0.118/0.007 for path, TF-IDF, blend, SIC-2, and random retrieval, with 99.2–99.6% target evaluability. The blend leads every method in 6/6 transitions. Here the evaluability rate is conditional on the ranking-time set,

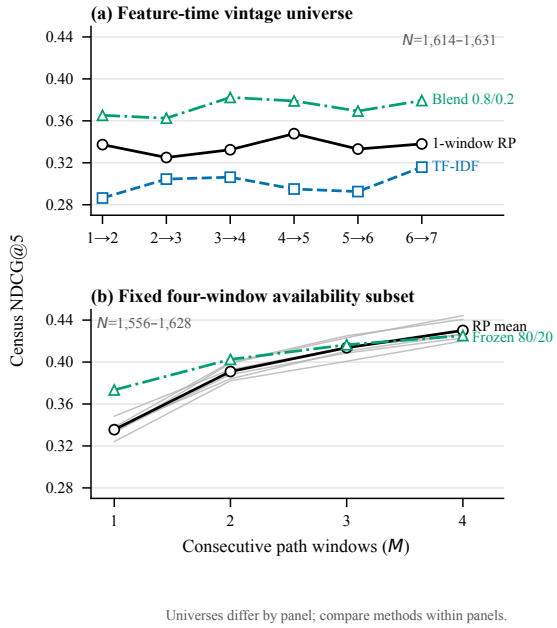


Figure 1: Modern walk-forward NDCG@5. (a) One-period path, TF-IDF, and frozen blend. (b) Path and the same blend by memory length on a fixed availability subset; thin lines are transitions and heavy lines are means. Fusion helps at $M = 1, 2$, but not $M = 4$. Panels use different universes.

not all firms a deployed system might encounter. The statistic uses the same W_{t+1} residual paths that define peer relevance. This audit measures fixed-list co-movement and evaluability only. It contains no analyst decisions, independently labeled shared drivers, or economic outcomes (Table 4).

Stale-return control. To separate the type of signal from its recency, we rank with the older W_{t-1} path while retaining W_t vintage TF-IDF and W_{t+1} relevance. On the identical N = 1,584–1,630 subset, the stale path reaches 0.328 ± 0.009 versus 0.301 ± 0.010 for TF-IDF and leads in 6/6; the current path reaches 0.335 ± 0.007. Filing age alone cannot explain the text gap.

5.3 Regime Robustness and Historical Extension

We next test whether the return path's advantage over standalone text methods extends across market regimes and the earlier 2020–2024 period.

We construct 13 additional non-overlapping calendar blocks (H1–H13; T = 62–86 trading days each, or approximately three to four months) spanning February 2020–April 2024, adjacent to W1 on the next-trading-day boundary. These periods contain N = 1,389–1,666 firms with complete returns and 1,264–1,547 with vintage text. Vintage 10-K coverage among firms with complete feature-period returns is 89.5–93.8% historically versus 82.3–84.1% in W1–W6, and the historical periods contain T = 62–86 trading days. The resulting mean (0.364) is therefore not directly comparable to Table 2 (0.336). NDCG@5 is higher in the high-volatility tercile.

Table 2: Six-transition walk-forward NDCG@5 (all queries; mean ± descriptive population SD). Groups use different eligibility rules. ‡ Transductive; spectral cosine is within 0.002 and omitted.

Universe	Method	WF NDCG@5
Full return	Return path	**0.339 ± 0.007**
	Spectral dot (rank 128)‡	0.336 ± 0.006
	SIC-2	0.213 ± 0.009
	Random	0.118 ± 0.006
Vintage 10-K	Frozen rank blend (80/20)	**0.373 ± 0.008**
	Return path	0.336 ± 0.007
	Vintage TF-IDF	0.300 ± 0.010
	Granite-r2 (stored excerpt)	0.279 ± 0.006
	Granite-r2 (512 tok.)	0.259 ± 0.006
	Vintage SEC-BERT	0.256 ± 0.009
	Vintage BGE	0.256 ± 0.007
	Vintage MiniLM	0.215 ± 0.006
Four-period history	Concatenated return path	**0.430 ± 0.009**

Table 3: Walk-forward NDCG@5 by volatility regime over 19 transitions spanning 2020–2026 (mean ± population SD; all queries per transition; LW shrinkage $\delta \in [0.62, 0.81]$ across the full 20-window panel). TF-IDF vocabulary and IDF weights are fitted separately on each window's corpus.

Regime	Return path	TF-IDF
All (19 trans.)	0.364 ± 0.033	0.318 ± 0.022
High-vol (7)	0.381 ± 0.038	0.333 ± 0.019
Medium-vol (6)	0.358 ± 0.030	0.317 ± 0.022
Low-vol (6)	0.348 ± 0.017	0.302 ± 0.012
Return path wins	18/19	

Because the regimes are descriptive and NDCG is normalized within query, we compare methods within transitions and do not interpret NDCG levels across periods. H13 ends the day preceding W1; the same FF5 residual, LW shrinkage, metric, and feature-time text rule are applied throughout. The extension spans the COVID-19 crash and the 2022 rate shock. TF-IDF weights are fitted per window. Each transition is labeled by the annualized volatility, during W_t, of the Fama–French market excess-return factor (Mkt-RF). The 33rd and 67th percentiles across all 20 windows label 7 high-, 6 medium-, and 6 low-volatility transitions. Because these thresholds use the full panel, the labels are descriptive rather than an ex-ante classifier. The return path leads in all three regimes (Table 3). We emphasize the observed within-transition ordering and treat the win counts descriptively. As a dependence check, odd- and even-indexed sets avoid direct reuse of adjacent periods. The return path wins 9/10 in the odd set and 9/9 in the even set. Securities and regimes remain shared, so the counts are not independent trials. Only H1→H2 reverses the ordering (TF-IDF leads by 0.039; Fig. 2); one transition cannot establish whether the crash-to-recovery change explains it.

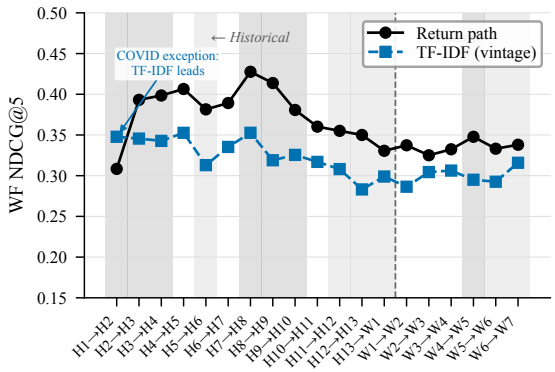


Figure 2: WF NDCG@5 per transition across the full 20-window, 19-transition evaluation (2020–2026; vintage-correct 10-K text; §5.3). Shading: high-volatility (darker), low-volatility (lighter), medium unshaded. Dashed line separates the historical extension (left) from W1–W7 (right). The return path leads in 18 of 19 transitions.

5.4 Robustness Checks

Ticker-syntax exclusion. A rule based on vendor ticker suffixes flags the same 118 possible preferred or non-primary share classes among firms eligible at each modern ranking date. After excluding them and refitting TF-IDF (N = 1,496–1,513), the return path remains ahead in 6/6 transitions: 0.324 ± 0.007 versus 0.301 ± 0.011, a gap of 0.023. Relative to the unrestricted universe, their mean changes are −0.012 and +0.001, respectively. The ticker rule is only a syntax check and does not identify common stocks authoritatively.

Residual-magnitude audit. Daily adjusted-price FF5 residuals include a few extreme magnitudes. Excluding any security whose maximum absolute raw residual exceeds 1.0 in decimal-return units (100 percentage points) in either the feature or target period removes 148 of 9,723 transition–security rows (1.52%; N = 1,587–1,604); after refitting TF-IDF and fully reranking, NDCG@5 is 0.340/0.305/0.378 for return path/TF-IDF/blend. A stricter threshold of 0.5 (50 percentage points) and feature-period screens at thresholds of 1.0 and 0.5 retain both method orderings in 6/6 transitions. Because screens that use either period inspect W_{t+1} and a large residual may be genuine, we use them only as retrospective data-quality checks.

Depth, weighting, and coverage. The return path exceeds TF-IDF at every tested depth $k \in \{1, 5, 10, 20\}$ (respective macro means 0.356/0.336/0.329/0.326 versus 0.317/0.300/0.288/0.280), and in 6/6 transitions at each depth. Giving recent days more weight with a fixed, RiskMetrics-style exponentially weighted moving average (EWMA) does not help: paths with 20- and 40-day half-lives attain 0.321 and 0.331, versus 0.336 unweighted. Requiring 10-K coverage only in W_t adds 0–3 names relative to requiring it in both W_t and W_{t+1}, and changes both return-path and TF-IDF means by less than 0.001.

Sensitivity to target dates. With rankings fixed, 1,000 circular block resamples per target period jointly resample contiguous 5- or 10-day blocks across all securities. The empirical 2.5th–97.5th percentiles give conditional 95% ranges of [0.031, 0.038] for return path minus TF-IDF and [0.029, 0.036] for blend minus return path. All six gaps remain positive in every replicate. The ranges describe conditional sensitivity and should not be read as confidence intervals.

Encoder context length. The median stored vintage Item 1 excerpt reaches the 8,000-character collection cap (85.9% reach it) and contains 1,511 Granite tokens; 94.6% exceed 512, while none exceed 8,192. Encoding the full stored excerpts raises Granite from 0.259 ± 0.006 at 512 tokens to 0.279 ± 0.006 and improves its score in all six transitions. Granite still trails TF-IDF and the return path in every transition. The test covers the stored excerpts, which may still truncate Item 1.

Historically trained nonlinear fusion. A histogram-based gradient-boosted pairwise ranker trained on twelve 2020–2024 transitions from return/text ranks and similarities, then frozen, attains 0.308 ± 0.009 and trails both the one-window path and fixed rank blend in 6/6 modern transitions. We include this model as a supervised diagnostic; the experiment does not cover the full range of learned peer models.

A second diagnostic replaces the fixed blend with a gated two-tower encoder: a temporal convolutional neural network (CNN) over residual paths, a frozen text projection, and a direct path for raw-return inputs. Trained and selected only on the same twelve historical transitions under a pre-registered continuation rule, it fails decisively: mean NDCG@5 is 0.377 versus 0.409 for the frozen blend across the six continuation tests, with 0/6 wins and little variation across seeds. As required, the model never accesses modern-transition targets. Neither trained model displaces the simple frozen blend under historical-only selection, supporting the residual path as a required baseline.

Contemporaneous text performance. Using the same eligible firms and frozen representations in each transition as the primary walk-forward comparison, but evaluating against W_t rather than W_{t+1} correlations, same-window NDCG@5 is 0.297 ± 0.007 for TF-IDF, 0.257 ± 0.006 for 512-token Granite-r2, and 0.213 ± 0.004 for MiniLM. The corresponding walk-forward values are 0.300 ± 0.010, 0.259 ± 0.006, and 0.215 ± 0.006; the largest unrounded gap is 0.0035. Within this diagnostic, contemporaneous evaluation does not close the text–return gap. Return availability in W_{t+1} is retained only to match the primary comparison universe.

Reproducibility. The paper is self-contained; code, window scripts, and hashed results are archived for release after review. Packages are pinned (`requirements.lock`); seeds, tie-breaking, and window parameters appear in Table 1. Price data come from Yahoo Finance via `yfinance`. Encoder weights are frozen public checkpoints. Raw price panels and per-security residual arrays are not redistributed; exact regeneration depends on Yahoo's adjusted history, which may be revised after acquisition. Frozen result files and their hashes are the authoritative numerical record.

6 Discussion

Return-path interpretation. The return path directly encodes sample correlation in one period. Its empirical value comes from how well the resulting neighbor ranking persists into the next period.

Table 4: Audit of lists fixed at ranking time (six equally weighted transitions). Lists are fixed before target returns; incomplete query–list pairs are dropped without reranking. For each list, r_{qB} correlates the query's raw W_{t+1} FF5 residual path with the daily equal-weight mean of its five peer paths; $\bar{r}_{qB}$ is its mean. Pos. is the share with $r_{qB} > 0$, and Eval. is the share with observed target returns, conditional on W_t return and vintage-text eligibility (N = 1,615–1,632). Positivity is directional, not a significance or economic-utility threshold.

Retrieval	$\bar{r}_{qB}$	Pos.	Eval.
Return path	0.208	74.1%	99.2%
TF-IDF	0.186	77.9%	99.4%
Frozen blend	**0.248**	**79.7%**	99.6%
SIC-2	0.118	71.6%	99.5%
Random	0.007	50.9%	99.4%

What the modalities capture. Filing text and residual returns encode different objects. Text primarily describes products and operations. Residual co-movement can reflect shared shocks, ownership, supply chains, or return drivers omitted from FF5 [1, 5, 9]. The comparison asks whether semantic similarity can recover this pattern when recent residual-return observations are available. The path is the stronger standalone signal. Combining text and returns helps with one or two periods of return history, but not four. Filing text without return history remains untested.

Pairwise alignment. Among firms eligible when the W1 ranking is formed (N = 1,614; 1,301,691 unordered pairs), Spearman correlation for the four dense encoders is only 0.029–0.046 against W1 residual correlation and 0.027–0.049 against W2. TF-IDF is 0.036/0.035, versus 1.000/0.103 for the return path (W1/W2). These finite-panel coefficients are descriptive because pairs share securities and target-return observations. They summarize global pairwise ordering, whereas NDCG@5 evaluates only five candidates per query.

Practical use. The motivating screen is refreshed only after the period's adjusted prices and public FF5 factors are available. An analyst would inspect its five candidates together with industry or valuation peers. The index stores 47–77 float32 values per security and requires a complete aligned path. For a new company or one with little history, a production system must abstain or use an untested fallback. Liquidity, stock-borrow availability, costs, and position constraints do not enter ranking. Production latency, factor-publication lag, data revisions, and analyst adoption are not measured here.

7 Limitations

Data and universe. The retrospectively assembled 2,200-ticker panel is not a point-in-time security master, so the historical extension is conditioned on later survival and news visibility. This may favor firms with stable exposures, but we cannot determine the direction of the resulting bias. Requiring complete returns in each period further favors liquid, surviving names; share classes and non-primary securities are not comprehensively removed. A ticker-syntax audit flags 118 possible preferred or non-primary names in the modern evaluation. The flag is not a definitive security classification. Results apply to this panel and should not be generalized to all U.S. common stocks. Non-U.S. markets are untested. Yahoo histories lack Center for Research in Security Prices (CRSP)-style delisting returns, security typing, and verified corporate-action adjustments. Residual-magnitude screens serve only as sensitivity checks and are not error labels.

Information sets and statistical scope. Because recent returns closely match the information needed to estimate future co-movement, the return representation has a deliberate advantage. The experiment tests whether that advantage persists; it cannot establish that the representation is optimal for other peer definitions. The primary modern NDCG evaluation defines its evaluated universe using complete returns in W_t and W_{t+1}. This makes it a retrospective benchmark, not a live backtest. A separate audit fixes rankings among firms eligible at W_t, then reports attrition without reranking (99.2–99.6% target evaluability). The rate is conditional on firms having returns and vintage text at ranking time. That selection universe covers 82.3–84.1% of the complete-return universe, not deployment coverage. The candidate set is still retrospective, so this is not a backtest built from a point-in-time security master. Adjacent transitions share one period, and queries share dates, candidates, and target matrices. These dependencies rule out IID inference. We report panel means, transition standard deviations, and win counts descriptively. Resampling target dates preserves common shocks but remains conditional on the observed regimes and does not establish population or cross-regime uncertainty.

Model and metric scope. The return path is unavailable for a newly listed security with no return history (a cold start) and otherwise requires a complete shared history. The experiments use one family of roughly quarterly windows, positive-correlation relevance, U.S. data, and no transaction-cost or portfolio optimization objective. NDCG is relative to each query's attainable ranking. Five names represent an illustrative analyst workload. We do not validate a minimum-correlation threshold, abstention rule, or utility–coverage curve relating performance to the share of queries served. Retrieval depth is tested at $k = 1, 5, 10, 20$, but temporal weighting is limited to two fixed half-lives and a descriptive concatenation of one to four periods. FF5-residual correlation is one peer definition among several. It does not reward same-issuer relationships that a valuation- or disclosure-based definition would rank first. Filing comparators are frozen off-the-shelf encoders. Because 85.9% of excerpts hit the 8,000-character collection cap, even the full-excerpt Granite run does not cover the complete section. MiniLM, BGE, and SEC-BERT use 128–512-token inputs (§5.4). Other current retrieval encoders, supervised filing models, risk factors (Item 1A), Management's Discussion and Analysis (MD&A), news, and Form S-1 registration filings, and vintage-aligned TNIC [22] remain open comparisons.

8 Conclusion

Semantic filing similarity is not a proxy for behavioral similarity in this experiment. Across 1,301,691 pairs, four dense encoders have Spearman correlations of only 0.029–0.046 with same-quarter residual co-movement and 0.027–0.049 with next-quarter co-movement.

Semantically similar firms, in other words, need not be behavioral peers. Yet text still helps at the top of the ranking when return history is short: the return path reaches NDCG@5 0.336 ± 0.007 and leads every standalone filing comparator, while a return–text blend reaches 0.373 ± 0.008 and improves all six modern transitions. Across 19 transitions the path leads TF-IDF in 18. The gain from text disappears with four periods of return history.

A separate audit freezes lists before target returns, leaves missing candidates unreplaced, and retains the blend's aggregate lead. It measures co-movement and evaluability, not exposure discovery, risk reduction, or analyst decisions.

For this return-defined task, recent return paths are a short, interpretable baseline when history exists. Other information settings require different encodings.

Generative-AI disclosure. OpenAI Codex and Anthropic Claude Code assisted with design review, writing, code refactoring, and result checks. The authors reviewed all text and code, reran the experiments, verified citations and numerical claims, and take responsibility for the work.

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